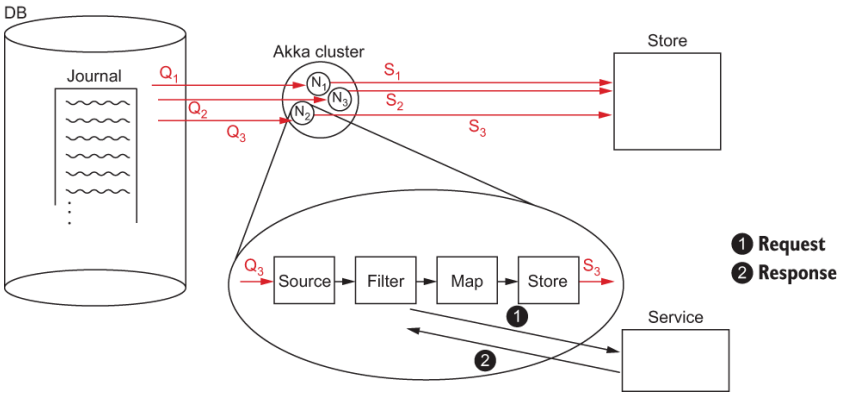


Figure 10.4 Pipes and filters with transformations

These streams are often seen in ETL (extract, transform, load) processes. In Akka, you can create similar processes with a *projection* that is a higher abstraction than a stream.

10.3 Projections

The epitome of a projection is mapping a sphere's points into a plane. It is a transformation from one form to another. A map, the end product of a mapping, is a projection, also called the *action* of its systematic representation. The same name is used for the product and the process.

An Akka projection can convert the events from the journal and move them to another type of storage, such as a database, a B-tree, or a list of logs like Kafka. One way or another, the data is transferred from the journal to another type of storage. You can read from Kafka and a database as source and write not only to Kafka or a database but also to a gRPC service.

Akka projections take care of reading the journal and offset for you. If the projection crashes, you don't have to worry about whether you've read an event when you restart. Projections can take care of that by setting offset semantics as effectively-once delivery—commonly called *exactly-once*. The other big advantage is scalability: projections have a built-in configuration that lets you set the level of parallelism you want. However, this is limited by the persistence entity tags.

There is a limitation with Akka projections using the Persistence Query module: the only method they implement from `ReadJournal` is `eventsByTag`. This is the only query you can ask for, but usually this is the query you are most interested in. This chapter covers the case of reading from the journal and writing to a database table.

NOTE The source code for this section is available at www.manning.com/books/akka-in-action-second-edition or https://github.com/franciscolopez sancho/akka-topics/tree/main/chapter10c. You can find the contents of any snippet or listing in the .scala file with the same name as the class, object, or trait.

To create a projection, you need two kinds of things:

- *Read side*—The selection of the source